Mechanical and Engineering

SEMESTER - I (Common for all B.Tech Courses)						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA 1XX	Linear Algebra: Theory and numerics	3	1.5	0	6
2		Introduction to Mechanical Engineering	3	0	0	6
3	CS101C	Computer Programming	3	0	2	9
4	ME201T	Engineering Mechanics	3	0	0	6
5		Formal and Technical communications	0	0	0	2
6	HS101T	Design Thiniking and Creativity	0	0	0	1
7	ME101L	hands on Engineering Laboratory	3	0	0	3
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
	First Semester Total Credits					35

Semester - II						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	MA XXX	Calculus and Vector Calculus	3	0	0	6
2		Enginering Gaphics	1.	0	3	6
3	ME206T	Therodynamics	3	0	0	6
4	CS101T	Data Structure and Algorithms	3	0	0	6
5	CS101L	Data Structure and Algorithms laboratory	3	0	0	3
6	ME204T	Manufacturing Processes-I	3	0	0	6
7		Engineering Metrology and Measurements	3	0	0	3
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				38

Semester III						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Introduction to Electrical and Electronics Circuits	3	0	0	6
2	ME 203	Fluid Mechanics	3	0	0	6
3	ME205T	Mechanics of Materials	3	0	0	6
4	ME202T	Engineering Materials	3	0	0	6
5	ME 2xxx	ODE's and Complex analysis	0	0	3	6
6		Machines Drowing and 3D modelling laboratory (New 5 Credit)	0	0	0	5
7	ME201L	manufacturing Processes and Metrology Laboratory	3	0	0	3
		Total Credits				38

Semester IV						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ME209T	Heat Transfer	3	0	0	6
2	ME211T	Manufacturing Processes-II	3	0	0	6
3	ME207T	Mechanical Measurements	3	0	0	6

4	ME304T	Theory of Machines	3	0	0	6
5	ME 3XXX	PDF's For Engineers	3	0	0	6
6	ME204L	Solid Mechanics Laboratory	3	0	0	3
7	ME202L	Fluid Mechnics Laboratory	3	0	0	3
8		Dynamics and Vibrations (new)	3	0	0	6
		Total Credits				42
Semester V						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ME306T	Design of Mechine Elements	3	0	0	6
2	EE205T	Introduction to Probability (1st half)	3	0	0	3
3	EE207T	Introduction to Data Analysis	3	0	0	3
4	HS201T	Economics	3	0	0	6
5	ME 3XXX	Numerical Methods for Engineers	3	0	0	6
6	ME301T	Applied Thermodynamics	3	0	0	6
7	ME203L	Mechanical Measurement Laboratory	3	0	0	3
8	ME205L	Heat Transfer Laboratory	3	0	0	3
9	ME206L	Manufacturing Processes Laboratory	3	0	0	3
		Total Credits				39
Semester VI						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE301T	Environmental Studies	3	0	0	6
2	ME208L	Control Systems and Laboratory	3	0	0	6
3		Thermal Engineering (New)	3	0	0	6
4		Operations Research and Industrial Engineering	3	0	0	6
5		Open Elective	3	0	0	6
6	ME302L	Applied Thermodynamics Laboratory	3	0	0	3
7	ME301L	Kinematics and Dynamics of Machinery Lab	3	0	0	3
8		Department Elective	3	0	0	6
		Total Credits				42
Semester VII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Department Elective	3	0	0	6
2		Department Elective	3	0	0	6
3		Open Elective	3	0	0	6
		Total Credits				18
Semester VIII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Department Elective	3	0	0	6
2		Open Elective	3	0	0	6
3		Open Elective	3	0	0	6
		Total Credits				18

Overal Credits Required (Minimum)	**270**

Mathematics and Computing

SEMESTER - I (Common for all B.Tech Courses)						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA108T	Calculus	3	1.5	0	9
2	PH101T	Quantum Physics and Applications	2	1	0	6
3	EE101T	Introduction to Electrical Systems and Electronic Circuits	3	0	0	6
4	ME101L	Hands on Engineering Laboratory	0	0	3	3
5	CS101C	Computer Programming	3	0	2	9
6	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
7	NO	NSO/NSS/NCC/NCA	0	0	2	2
	Total Credits					35

Semester - II						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	MA109T	Linear Algebra and Diffrential Equations	3	1.5	0	9
2	CS101T	Data Structures and Algorithms	3	0	0	6
3	CS101L	Data Structures and Algorithms Laboratory	0	0	3	3
4	EE103T	Digital Systems				6
5	EE XXX	Digital Systems Lab				3
6	CS202T	Discrete Structures	3	0	0	6
7	CE301T	Environmental studies	3	0	0	6
8	NO	NSO/NSS/NCA/NCC	0	0	0	2
		Total Credits				39+2

Semester III						
S.No	**Course code**	**Course name**	**L**	**T**	**P**	**C**
1	MA401T	Introduction to probability Theory	3	1.5	0	9
2	CS201T	Automata Theory	3	15	0	9
3	CS407T	Mathematics for Data Science				6
4	CS203T	Design and Analysis of Algorithms	3	0	0	6
5	CS201C	Software Systems Lab	2	0	2	6
Total credits						36

Semester IV						
S.No	**Course code**	**Course name**	**L**	**T**	**P**	**C**
1	MA402T	Statistics	3	0	0	6
2	MA302T	Real Analysis	2	1	0	6
3	CS204T	Artificial Intelligence	3	0	0	6

4	CS201L	Artificial Intelligence Laboratory	0	0	3	3
5	CS301L	Computer Architecture Laboratory	0	0	3	3
6	CS302T	Computer Architecture	3	0	0	6
7	MA 210	Statistics	3	0	0	6
Total credits						36

Semester V

S.No	Course code	Course name	L	T	P	C
1	MA407T	Stochastic Models	3	0	0	6
2	MA404T	Introduction to Mathematical Finance I	3	0	0	6
3	CS403T	Introduction to Graph Theory	3	0	0	6
4		Elective				6
5		Elective				6
6	HS 201	Economics	3	0	0	6
Total credits						36

Semester VI

S.No	Course code	Course name	L	T	P	C
1	MA303T	Group Theory	3	1	0	6
2	MA408T	Introduction to Mathematical Finance 2	3	0	0	6
3	MA107T	Advanced Linear Algebra	3	1	0	6
4		Elective				6
5		Elective				6
	MA409T	numerical Linear Algebra	3	0	0	6
Total credits						36

Semester VII

S.No	Course Code	Course Name	L	T	P	C
1		Elective				6
2		Elective				6
3		Elective				6
4		Elective				6
Total credits						24

Semester VIII

S.No	Course Code	Course Name	L	T	P	C
1		Elective				6
2		Elective				6
3		Elective				6
Overal Credits Required (Minimum)						18

	Overal Credits Required (Minimum)	**262**

Engineering Physics

SEMESTER - I (Common for all B.Tech Courses)						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA108T	Calculus	3	1..	0	9
2	PH101T	Quantum Physics and Applications	2	1	0	6
3	CY101T	Fundamental Concepts &Applications Of Chemistry	3	0	0	6
4	EE101T	Introduction to electrical systems and Electronics	3	0	0	6
5	PH101L	Hands on Science Laboratory - 1	0	0	3	3
6	CS101C	Computer Programming	3	0	3	9
7	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
	First Semester Total Credits					41

SEMESTER - II						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CS101T	Data Structures and Algorithms	3	0	0	6
2	ME206T	Thermodynamics	3	0	0	6
3	MA109T	Linear Algebra and Diffrential Equations	3	1..	0	9
4	PH102T	Electricity and Magnetism	2	1	0	6
5	ME101L	Hands on Engineering Laboratory	0	0	3	3
6	ME102T	Engineering Graphics	1..	0	3	6
7	CS101L	Data Structure & Algorithms Lab	0	0	3	3
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
	Total Credits					41

SEMESTER - III						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH302T	Classical Mechanics	2	1	0	6
2	PH303T	Quantum Mechanics - I	2	1	0	6
3	ME201T	Engineering Machines	3	0	0	6
4	EE204T	Signals and Systems	2	1	0	6
5	EE205T	Introduction to Probability (Pre mid-sem)	3	0	0	3
6	EE207T	Data Analysis (Post MId Sem)	2	1	0	3
7	HS 201	Economics	2	1	0	6
8	PH201L	Introductory Physics Laboratory	0	0	3	3
	Third Semester Total Credits					39

SEMESTER - IV						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH305T	Statistical Physics	2	1	0	6
2	EE103T	Digital Systems	2	1	0	6

3	CS302T	Computer Architecture	2	1	0	6
4	PH304T	Mathematical Physics	2	1	0	6
5	PH301L	General Physics Laboratory	0	0	3	3
6	EE101L	Digital Circuits Laboratory	0	0	3	3
7	CS301L	Computer Architecture Lab	0	0	3	3
	Fourth Semester Total Credits					33

SEMESTER - V						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH301T	Electrodynamics	2	1	0	6
2	PH420T	Electronics	3	0	0	6
3	EE302C	Fundamentals of Digital Signal Processing	2	0	2	6
4	ME203T	Fluid Mechanics	3	0	0	6
5	PH402L	Electronics Loboratory	0	0	3	3
6	PH401L	Advanced Physics Laboratory	0	0	3	3
7		Institute Elective-				6
	Fifth Semester Total Credits					36

SEMESTER - VI						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH401T	Quantum Mechanics-II	2	1	0	6
2	PH402T	Condensed Matter Physics	2	1	0	6
3	CE301T	Environmental Studies	3	0	0	6
4	CS204T	Artificial Inteligence	3	0	0	6
5	--	Institute Elective-2/RnD Project-I or II				6
6	ME207L	Mechanics and Measurement Lab	0	0	3	3
7	CS201L	Artificial Inteligence Lab	0	0	3	3
	Sixth Semester Total Credits					36

SEMESTER - VII						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CS601S	Seminar	0	0	4	4
2		Institute Elective-3				6
3	PH501C	Numerical Methods	2	0	2	6
4		Institute Elective-4/BTP-I				6
	Seventh Semester Total Credits					22

SEMESTER -VIII						
Sl. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Institute Elective – 5	2	1	0	6
2		Institute Elective – 6	2	1	0	6
3		Institute Elective – 7/ BTP-II	2	1	0	6
	Eighth Semester Total Credits					18

	Overal Credits Required (Minimum)	257/266

Electrical and Electronics Engineering

Semester - II

S. No	Course Code	Course Name	L	T	P	C
1	MA XXX	Linear Algebra and Diffrential equations	3	0	0	9
2	EE 204	Digital Systems	2	1	0	6
3	EE101L	Digital Circuits Lab	0	0	3	3
4	EE104T	Network Theory	2	1	0	6
5	EE105T	Electronic Devices (2nd Half)	3	0	0	3
6	ME101C	Engineering Graphics Lab	1	0	3	5
7	ME101L	Hands on Engineering Lab	0	0	3	3
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				**37**

Semester III

S.No	Course Code	Course Name	L	T	P	C
1	MA110T	Complex Analysis and Differential Equations II	3	0	0	9
2	EE205T	Introduction to Probability	3	0	0	3
3	EE207T	Data Analysis (2nd Half)	3	0	0	3
4	EE204T	Signals and Systems	2	1	0	6
5	EE201T	Introduction to Analog Circuits (1st Half)	3	0	0	3
6	EE201L	Devices and Circuit Lab	0	0	3	3
7	PH102T	Electricity and Magnetism			6	6
8	HS201T	Economics	3	0	0	6
		Total Credits				39

Semester - IV

S. No	Course Code	Course Name	L	T	P	C
1	EE 239	Control Systems Engineering	3	0	0	6
2	EE203L	Control Systems Engineering Lab	0	0	3	3
3	EE209T	Introduction to modern communication systems	2	0	2	6
4	EE202T	Introduction to Electrical Machines (1st Half)	3	0	0	3
5	EE206T	Introduction to Power Systems (2nd Half)	2	0	2	3
6	EE203T	Introduction to Power Electronics (1st Half)	3	0	0	3
7	EE202L	Electrical Mechines and Power Electronics Lab	0	0	3	3
8	CS101T	Data Structures and Algorithms	3	0	0	6
9	CS602L	Data Structures and Algorithms Lab	0	0	3	3
		Total Credits				36

Semester - IV

S. No	Course Code	Course Name	L	T	P	C
1	EE208T	Control Systems Engineering	3	0	0	6

2	EE203L	Control Systems Engineering Lab	0	0	3	3
3	EE209T	Introduction to modern communication systems	2	0	2	6
4	EE202T	Introduction to Electrical Machines (1st Half)	3	0	0	3
5	EE206T	Introduction to Power Systems (2nd Half)	2	0	2	3
6	EE203T	Introduction to Power Electronics (1st Half)	3	0	0	3
7	EE202L	Electrical Mechines and Power Electronics Lab	0	0	3	3
8	CS101T	Data Structures and Algorithms	3	0	0	6
9	CS101L	Data Structures and Algorithms Lab	0	0	3	3
		Total Credits				36

Semester V						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	MA412T	Numerical Analysis	2	1	0	6
2	EE301T	Microprocessors and Microcontrollers	3	0	0	6
3	EE302L	Microprocessors and Microcontrollers Lab	0	0	3	3
4	EE302C	Fundamentals of Digital Signal Processing	2	0	2	6
5		Electives				18
	Total Credits					39

Semester VI						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE301T	Environmental Studies	3	0	0	6
2	EE301L	Electronic Design Laboratory	3	0	0	6
3		Elective Courses				24
	EE301O	Technical Writing				2
		Total Credits				38

Semester VII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Electives				30

Semester VIII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Electives				18

	Overal Credits Required (Minimum)	**269**

The courses that the students take in the third and fourth years define the programme. In semester 5 (Elective_18) and semester 6 (Elective _24)

Electronics and Communication Engineering

SEMESTER - I (Common for all B.Tech Courses)

S.No	C. Code	Course	L	T	P	C
1	MA108T	Calculus	3	1.5	0	9
2	EE101T	Introduction to Electrical Systems and Electronics	3	0	0	6
3	CS101C	Computer Programming	3	0	2	8
4	PH101T	Quantum Physics and Applications	2	1	0	6
5	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
6	CC	NSO/NSS/NCC/NCA	0	0	2	2
7	EE101O	Formal Communications	1	0	0	2
	First Semester Total Credits					33

Semester - II

S. No	Course Code	Course Name	L	T	P	C
1	MA109T	Linear Algebra and Differential Equations	3	1.5	0	9
2	EE103T	Digital Systems	2	1	0	6
3	EE101L	Digital Circuits Lab	0	0	3	3
4	EE104T	Network Theory	2	1	0	6
5	EE105T	Electronic Devices (2nd Half)	3	0	0	3
6	ME101C	Engineeing Graphics Laboratory	1	0	3	5
7	ME101L	Hands on Engineering Laboratory	0	0	3	3
8	CC	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				**37**

Semester III

S.No	Course Code	Course Name	L	T	P	C
1	MA110T	Complex Analysis and Differential Equations II	3	0	0	9
2	EE205T	Introduction to Probability (1st Half)	3	0	0	3
3	EE207T	Data Analysis (2nd Half)	3	0	0	3
4	EE204T	Signals and Systems	2	1	0	6
5	EE201T	Introduction to Analog Circuits (1st Half)	3	0	0	3
6	EE201L	Devices and Circuit Lab	0	0	3	3
7	PH102T	Electricity and Magnetism			6	6
8	HS201T	Economics	3	0	0	6
		Total Credits				39

Semester - IV

S. No	Course Code	Course Name	L	T	P	C
1	EE 239	Control Systems Engineering	3	0	0	6
2	EE203L	Control Systems Engineering Lab	0	0	3	3
3	EE209T	Introduction to modern communication systems	2	0	2	6

4	EE202T	Introduction to Electrical Machines (1st Half)	3	0	0	3
5	EE206T	Introduction to Power Systems (2nd Half)	2	0	2	3
6	EE203T	Introduction to Power Electronics (1st Half)	3	0	0	3
7	EE202L	Electrical Mechines and Power Electronics Lab	0	0	3	3
8	CS101T	Data Structures and Algorithms	3	0	0	6
9	CS602L	Data Structures and Algorithms Lab	0	0	3	3
		Total Credits				36

Semester V

S.No	Course Code	Course Name	L	T	P	C
1	MA412T	Numerical Analysis	2	1	0	6
2	EE301T	Microprocessors and Microcontrollers	3	0	0	6
3	EE302L	Microprocessors and Microcontrollers Lab	0	0	3	3
4	EE302C	Fundamentals of Digital Signal Processing	2	0	2	6
5		Electives				18
	Total Credits					39

Semester VI

S.No	Course Code	Course Name	L	T	P	C
1	CE301T	Environmental Studies	3	0	0	6
2	EE301L	Electronic Design Laboratory	3	0	0	6
3		Elective Courses				24
	EE301O	Technical Writing				2
		Total Credits				38

Semester VII

S.No	Course Code	Course Name	L	T	P	C
1		Electives				30

Semester VIII

S.No	Course Code	Course Name	L	T	P	C
1		Electives				18

	Overal Credits Required (Minimum)	**269**

The courses that the students take in the third and fourth years define the programme. In semester 5 (Elective_18) and semester 6 (Elective _24)

Electrical Engineering

Semester - I (Common for all B.Tech Courses)

S.No	C. Code	Course	L	T	P	C
1	MA 101	Calculus	3	1	0	8
2	PH 101	Quantum Physics and Applications	2	1	0	6
3	CH 102	Fundamental Concepts and Applications of Chemistr	3	0	0	6
4	BB 103	Introduction to Modern biology	3	0	0	6
5	PH 113	Hands on Science Laboratory - I	0	0	3	3
6	CS 101	Computer Programming	3	0	2	8
7	HS 103	Introduction to Fine Arts	0	0	1	PP/NP
8	HS 106	Design Thinking and Creativity	1	0	0	PP/NP
9	NO107/NO105	NSO/NSS	0	0	2	2
	First Semester Total Credits					37

Semester - II

S. No	Course Code	Course Name	L	T	P	C
1	MA 102	Linear Algebra (1st Half)	3	1	0	4
2	MA 103	Differential Equations - I (2nd Half)	3	1	0	4
3	ME 111	Engineering Graphics Laboratory	1	0	3	5
4	EE 101	Introduction to Electrical Systems and Electronics	3	0	0	6
5	CS 201	Data Structures and Algorithms	3	0	0	6
6	CS 211	Data Structures and Algorithms Laboratory	0	0	3	3
7	ME 113	Hands on Engineering Laboratory	0	0	3	3
8	PH 102	Electricity and magnetism	2	1	0	6
9	NO 105/ NO 107	National Sports Organization (NSO)/National Service Scheme (NSS)	0	0	2	2
		Total Credits				37

Semester III

S.No	Course Code	Course Name	L	T	P	C
1	EE 221	Introduction to Probability (1st Half)	3	0	0	3
2	EE 227	Data Analysis (2nd Half)	3	0	0	3
3	EE 229	Electronic Devices (1st Half)	3	0	0	3
4	EE 202	Introduction to Analog Circuits (2nd Half)	3	0	0	3
5	EE 205	Network Theory	2	1	0	6
6	EE 210	Signals and Systems	2	1	0	6
7	MA 201	Complex Analysis (1st Half)	3	1	0	4
8	MA 203	Differential Equations II (2nd Half)	3	1	0	4

9	HS 201	Economics	3	0	0	6
		Total Credits				38

Semester - IV						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	EE 206	Introduction to Electrical Machines (1st Half)	3	0	0	3
2	EE 209	Introduction to Power Electronics (2nd Half)	3	0	0	3
3	EE 208	Engineering Electromagnetics (1st Half)	2	1	0	3
4	EE 223	Introduction to Power Systems (2nd Half)	2	0	2	3
5	EE 232	Introduction to Communication Systems (1st Half)	3	1	0	3
6	EE 216	Comunications Lab (2nd Half)	0	0	4	2
7	EE 204	Digital Systems	3	0	0	6
8	EE 214	Digital Circuits Lab	0	0	3	3
9	EE 226	Control Systems and Laboratory	2	0	2	6
10	EE 212	Devices and Circuits Lab	0	0	3	3
		Total Credits				35

Semester V						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	EE301T	Microprocessors and Microcontrollers	3	0	0	6
2	EE302C	Digital Signal Processing	3	0	0	3
3	EE304L	Digital Signal Processing Lab	0	0	4	2
4	EE302L	Microprocessors and Microcontrollers Lab	0	0	3	3
5	EE202L	Electrical Mechines and Power Electronics Lab	0	0	3	3
6		HSS Elective	3	0	0	6
7		Electives				12
	Total Credits					35

Semester VI						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE301T	Environmental Studies	3	0	0	6
2	EE301L	Electronics Design Lab	3	0	0	6
3		Elective Courses				24
		Total Credits				36

Semester VII & VIII		
1. Student has to earn 36 credits in the fourth year.		
2. Student may choose to earn zero or 6 or 12 credits though BTP/co-op project.		
3. The BTP/co-op may be split in two semesters (6 credits per semester).		
4. The remaining credits should be earned through Institute Elective.		
	Overal Credits Required (Minimum)	254

Computer Science and Engineering

SEMESTER - I (Common for all B.Tech Courses)

S.No	C. Code	Course	L	T	P	C
1	MA108T	Calculus	3	1.5	0	9
2	EE101T	Introduction to Electrical Systems and Electronics	3	0	0	6
3	CH101T	Fundamental Concepts and Applications of Chemistry	3	0	0	6
5	ME101L	Hands on Engineering Lab	0	0	3	3
6	CS101C	Computer Programming	3	0	2	9
7	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
8	CC	NSO/NSS/NCC/NCA	0	0	2	2
	First Semester Total Credits					35

Semester - II

S. No	Course Code	Course Name	L	T	P	C
1	MA109T	Linear Algebra and Differential Equations	3	1.5	0	9
2	EE103T	Digital Systems	3	0	0	6
3	EE101L	Digital Circuits Lab	0	0	3	3
4	CS101T	Data Structures and Algorithms	3	0	0	6
5	CS202T	Discrete Structures	3	0	0	6
6	CS101L	Data Structures and Algorithms Laboratory	0	0	3	3
7	CC	NSO/NSS/NCC/NCA	1	0	0	2
		Total Credits				35

Semester III

S.No	Course code	Course name	L	T	P	C
1	CS 205	Design and analysis of algorithms	3	0	0	6
2	EE205T	Introduction to Probability (1st Half)	3	0	0	3
3	EE207T	Data Analysis (2nd Half)	3	0	0	3
4	CS201T	Automata Theory	3	1.5	0	9
5	CS201C	Software Systems Lab	3	0	3	9
6	PH101T	Quantum Physics and Applications	3	0	0	6
Total credits						36

Semester IV

S.No	Course code	Course name	L	T	P	C
1	CS204T	Artificial intelligence	3	0	0	6
2	CS501C	Computer Architecture	3	0	0	6
3	CE301T	Environmental Studies	3	0	0	6
4	CS205T	Data Bases and Information Systems	3	0	0	6
5	CS201L	Artificial Intelligence lab	0	0	3	3
6	CS202L	Data Bases and Information Systems Laboratory	0	0	3	3
7	CS301L	Computer Architecture lab	0	0	3	3

Total credits						33

Semester V

S.No	Course code	Course name	L	T	P	C
1	CS303T	Operating systems	3	0	0	6
2	CS305T	Computer Networks	3	0	0	6
3	CS303L	Computer Networks lab	0	0	3	3
4	CS302L	Operating systems lab	0	0	3	3
5	BB101T	Introduction to Modern Biology	3	0	0	6
6		Electives	3	0	0	6
	HS201T	Economics	3	0	0	6
Total credits						36

Semester VI

S.No	Course code	Course name	L	T	P	C
1	CS304T	Compilers	3	0	0	6
2	CS304L	Compilers lab	0	0	3	3
3	EE301O	Technical Writing (P/NP) No Graded	1	0	0	2
4		Electives				24
5	EE101O	Formal Communication No Graded	1	0	0	2
Total credits						37

Semester VII

S.No	Course Code	Course Name	L	T	P	C
1		Elective IV				30
Total credits						24

Semester VIII

S.No	Course Code	Course Name	L	T	P	C
1		Elective				6
Total credits						18

	Overal Credits Required (Minimum)	**254**

Civil and Infrastructure Engineering 2026

SEMESTER - I (Common for all B.Tech Courses)						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	CH101T	Fundamental Concepts & Applications of Chemistry	3	0	0	6
2	MA108T	Calculus	3	1.5	0	9
3	ME201T	Engineering Mechanics	2	1	0	6
4	PH101T	Quantum Physics and Applications	2	1	0	6
5	CS101C	Computer Programming	3	0	3	9
6	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
7	NO 101/NO 103	National Sports Organization	0	0	2	2
		Total Credits				**38**

Semester II						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**Total**
1	CE101T	Introduction to Civil Engineering	2	1	0	6
2	CE301T	Environmental Studies	3	0	0	6
3	MA109T	Linear Algebra and Differential Equations	3	1.5	0	9
4	ME101C	Engineering Graphics Lab	1	0	3	5
5	CS101T	Data Structures and Algorithms	3	0	0	6
6	CS101L	Data Structures and Algorithms Laboratory	0	0	3	3
7	ME101L	Hands-on Engineering Lab	0	0	3	3
8	NO107/ NO105/ N	NSO/NSS/NCC/NCA	0	0	2	2
		Total Number of Credits				**40**

Semester III						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE205T	Building Materials and Construction	3	0	0	6
2	CE206T	Sustainable Infrastructure Planning	3	0	0	6
3	ME203T	Fluid Mechanics	3	0	0	6
4	ME205T	Mechanics of Materials	2	1	0	6
5	HS201T	Economics	2	1	0	6
6	CE204L	Construction Materials Laboratory	0	0	3	3
7	CE203L	Building Drawing Practice	0	0	3	3
		Total Credits				**36**

Semester IV						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE202T	Surveying and Geomatics	2	1	0	6
2	CE202L	Surveying and Geomatics Laboratory	0	0	3	3
3	CE203T	Structural Analysis	2	1	0	6
4	ME204L	Solid Mechanics Laboratory	0	0	3	3
5	ME202L	Fluid Mechanics Laboratory	0	0	3	3
6	CE207T	Hydraulics and hydraulic Machinery	3	0	0	6
7	CE204T	Water Resources Engineering	3	0	0	6

		Total Credits				**33**
Semester V						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE302T	Design of Concrete Structures	2	1	0	6
2	CE303T	Geotechnical Engineering	2	1	0	6
3	CE304T	Transportation Engineering	2	1	0	6
4	CE302L	Transportation Engineering Laboratory	0	0	3	3
5	CE303L	Geotechnical Engineering Laboratory	0	0	3	3
6	CE305T	Environmental Engineering	2	1	0	6
7	CE304L	Environmental Engineering Laboratory	0	0	3	3
		Total Credits				**33**
Semester VI						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE306T	Design of Steel Structures	2	1	0	6
2	CE309T	Construction Engineering and management	3	0	0	6
3	CE307T	Foundation Engineering	2	1	0	6
4		Institute Elective-I/R&D Project in Civil Engineering	2	1	0	6
5	CE306L	Sensors Technologies & Instrumentation in Civil Engineering	0	0	3	3
6		Institute Elective-II	2	1	0	6
		Total Credits				**33**
Semester VII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE402C	Civil and Infrastructure Engineering Design	1	1	2	6
2	CE401P	BTP-I/Intensive BTP-1*	0	0	6/9	6/9
3	CE308T	Estimation and Costing in civil Engineering	2	1	0	6
4		Institute Elective-III/ Program Elective-I	2	1	0	6
5		Institute Elective-IV	2	1	0	6
		Total Credits				**30/33**
Semester VIII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE402P	BTP-II/Program Elective-II	0	0	6/9	6/9
2		Institute Elective-V/Program Elective-II	0	0	0	6
		Total Credits				**12/15**
	Overal Credits Required (Minimum)					**255/261**

Chemical and Biochemical Engineering

SEMESTER - I (Common for all B.Tech Courses)						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA108T	Calculus	3	1.5	0	9
2	PH101T	Quantum Physics and Applications	2	1	0	6
3	PH101L	Hands on Science Laboratory – I	0	0	3	3
4	CS101C	Computer Programming	3	0	3	9
5	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
6	CH101T	Fundamental Concepts & Applications of Chemistry	3	0	0	6
7	BB101T	Introduction to Modern Biology	3	0	0	6
8	CC	NCC/NSO/NCA/NSS	0	0	2	2
		Total Credits				**41**

Semester II						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	MA109T	Linear Algebra and Differential Equations	3	1.5	0	9
2	ME101C	Engineering Graphics Lab	1	0	3	6
3	ME101L	Hands on Engineering Lab	0	0	3	3
4	CL101T	Introduction to Chemical Engineering	3	0	0	6
5	BB201T	Biomolecules	2	1	0	6
6	CH201T	Organic Chemisrty	3	0	0	3
7	CC	NCC/NSO/NCA/NSS	0	0	2	2
		Total Credits				**35**

Semester III						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	EE205T	Introduction to Probability	3	0	0	6
2	EE207T	Data analysis	3	0	0	3
3	CL204T	Fluid Mechanics and Mechanical Operations	2	1	0	6
4	CL205T	Heat Transfer for Chemical Engineers	3	0	0	6
5	CL203T	Introduction to Chemical Engineering Thermodynamics	3	0	0	6
6	ME201T	Engineering Mechanics	2	1	0	6
7	BB401T	Basics of Cell Biology and Genetics	3	0	0	6
		Total Credits				**36**

Semester IV						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CS204T	Artificial Intelligence	3	0	0	6
2	CS201L	Artificial Intelligence lab	0	0	3	3

3	CL206T	Materials Science for Chemical Engineers	3	0	0	6
4	BB403T	Biophysics	3	0	0	3
5	BB201L	Biology Lab I	0	0	3	3
6	CL201L	Chemical Engineering lab 1 (Thermodynamics and fluid mechanics)	0	0	3	3
7	CL201T	Reaction Engineering	3	0	0	6
8	CL202T	Mass Transfer	3	0	0	6
		Total Credits				**36**

Semester V						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CL304T	Applications of Mass Transfer	3	0	0	6
2	CL308T	Advanced Chemical Reaction Engineering	3	0	0	6
3	CL307T	Process Dynamics and Control	3	0	0	6
4	CL302T	Process Equipment Design and Economics	3	0	0	6
5	CL301L	Chemical Engineering Lab II	0	0	3	3
6	HS201T	Economics	3	0	0	6
		Total Credits				**33**

Semester VI						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CL301T	Introduction to Transport Phenomena	3	0	0	6
2	CL301C	Numerical Method in Chemical Engineering	3	0	0	6
3	CL302L	Chemical Engineering Lab III	0	0	3	3
4		Program Elective-I	3	0	0	6
5	CL306T	AI in Chemical Engineering	3	0	0	6
6	CL303L	Chemical Engineering Lab IV	0	0	3	3
7	CE301T	Environmental Studies	3	0	0	6
		Total Credits				**36**

Semester VII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CL405T	Bio Process Engineering	3	0	0	6
2	CL401S	Scientific Presentation	0	0	3	3
3		Program Elective-II	3	0	0	6
4		Program Elective-III/BTP-I	3	0	0	6
5		HSS Elective-I	3	0	0	6
		Total Credits				**27**

Semester VIII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Program Elective-IV	3	0	0	6
2		Program Elective-V/BTP-II	3	0	0	6
3		Institute Elective-I	3	0	0	6
4		HSS Elective-II	3	0	0	6

		Total Credits		**24**
	Overal Credits Required (Minimum)			**268**
	Program Electives	5		
	Institute Electives	1		
	HSS Electives	2	**268**	